



Spanwise layout optimization of aerodynamic countermeasures for multi-mode vortex-induced vibration control on long-span bridges

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ABSTRACT

Aerodynamic countermeasures are widely used to control vortex-induced vibration (VIV) on long-span bridges. However, due to the lack of appropriate three-dimensional (3-D) VIV analysis approach, these countermeasures are usually installed along the entire span of the bridge deck, leading to excessive costs. This study proposes an optimization method for the spanwise layout of aerodynamic countermeasures, aiming to achieve an economical control scheme for multi-mode VIV. An improved mode-by-mode 3-D VIV analysis approach is developed based on the generalized polynomial model to predict the VIV responses of the bridge with different spanwise layouts of aerodynamic countermeasures. The genetic algorithm (GA) is then employed to search for the most economical layout scheme with constraints of limited multi-mode VIV responses. Two alternative types of search spaces for the optimization are presented. The effectiveness of this optimization method is demonstrated on a long-span suspension bridge in conjunction with finite element simulation and wind tunnel tests. The results show a noteworthy reduction in the use of aerodynamic countermeasures, thus highlighting the necessity of countermeasure layout optimization.

1. Introduction

Vortex-induced vibration (VIV) is one of the major issues in long-span bridge vibration. This type of oscillation occurs when the dominant vortex shedding frequency approaches the natural frequency of the bridge, which is known as the “lock-in” phenomenon. Large-amplitude VIV is prone to occur at relatively low wind speeds and thus threatens the serviceability and fatigue life of bridges. The occurrences of vertical VIV have been reported on several long-span bridges, such as Rio-Niterói Bridge (Battista and Pfeil, 2000), Storebælt Suspension Bridge (Larsen et al., 2000), Trans-Tokyo Bay Crossing Bridge (Fujino and Yoshida, 2002), Second Severn Crossing Bridge (Macdonald et al., 2002) and Xihoumen Bridge (Li et al., 2011). More recently, the VIV on Humen Bridge (Ge et al., 2022) in China has attracted extensive attention. These in-situ observations show that VIV occurs in one mode of the bridge at a time, but vibrations in different modes will be excited as the wind speed changes, which is because long-span bridges are flexible structures with low and crowded modal frequencies. Therefore, it is necessary to precisely evaluate the multi-mode VIV performance of long-span bridges

and mitigate any undesirable vibration with effective countermeasures.

Extensive types of countermeasures have been proposed to control VIV on bridge decks, which can be roughly categorized into the aerodynamic and mechanical ones (Fujino and Siringoringo, 2013). Compared to the mechanical type that employs vibration control devices such as the tuned mass damper, the aerodynamic countermeasure is more widely adopted in practice due to its stable control effect and convenient operation. This kind of countermeasure aims to directly reduce the wind-induced force and thus the structural VIV response by optimizing the outer shape of the bridge deck (Zhao et al., 2019). In addition to the shape optimum design of the original bridge deck, some aerodynamic appendages such as guide vanes (Larsen et al., 2000, 2008), baffles (Macdonald et al., 2002), wind barriers (Li et al., 2011), and flaps (Ge et al., 2022), are needed to enhance the control efficiency for VIV. Due to the lack of theoretical guidance, aerodynamic countermeasures for a specific bridge deck are usually determined through trial and error in wind tunnel tests of the deck sectional model. The countermeasures derived from the two-dimensional (2-D) section model test should then be implemented along the whole span of the prototype

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bridge deck. However, the installation of aerodynamic appendages throughout the entire span could result in prohibitively high construction and maintenance costs.

According to previous studies (Ehsan and Scanlan, 1990; Zhu et al., 2013, 2017a), the aerodynamic force acting on a bridge deck at vortex lock-in is dominated by the self-excited force component, including the aerodynamic damping and stiffness forces. The magnitude of this component increases with the vibration amplitude of the bridge deck, making the aerodynamic force more significant in areas with larger deck response. As a result, installing countermeasures at the spanwise locations with small bridge deck response (e.g., those near the bridge bearings) has a very limited contribution to mitigating VIV on the bridge deck. Therefore, it is possible to reduce the use of aerodynamic countermeasures by optimizing their installation positions along the span. However, it should be noted that the countermeasure spanwise layout optimization discussed here assumes that the flow field around the whole span of bridge deck is an extension of the transverse 2-D flow and neglects the influence of spanwise flow. Accordingly, this idea differs from the three-dimensional (3-D) control method (El-Gammal et al., 2007; Xin et al., 2018; Zhang et al., 2022), wherein aerodynamic countermeasures are non-uniformly placed along the span to generate spanwise perturbation and suppress the regular vortex shedding.

Prediction of VIV response for bridges with multiple-cross-section deck is crucial for optimizing the spanwise layout of aerodynamic countermeasures, which should be efficient and accurate enough to evaluate the multi-mode VIV performance of bridges with different countermeasure layout schemes. As VIV usually occurs in a single mode of the bridge at a time, the mode-by-mode approach is commonly used to analyse the VIV response (Ehsan and Scanlan, 1990; Pan et al., 2022; Xu et al., 2018; Zhang et al., 2020; Zhu et al., 2022). This analytical approach requires an appropriate mathematical model to simulate the 2-D aerodynamic force on bridge deck section, which is then utilized to the full span with consideration of spanwise correlation and structural mode shape. The accuracy of the analysis outcomes is highly reliant on the employed aerodynamic force model. Most of the mode-by-mode 3-D VIV analyses focused on bridges with uniform deck (Ehsan and Scanlan, 1990; Xu et al., 2018; Zhang et al., 2020; Zhu et al., 2022). Pan et al. (2022) also used this approach to compute VIV response of a bridge with multiple-cross-section deck. In Pan's study, Scanlan's empirical nonlinear model (Ehsan and Scanlan, 1990) and a linear aerodynamic damping model were respectively employed for the VIV-prone and VIV-insensitive bridge deck sections. However, these lower-order force models are unreliable for the prediction of VIV response due to the high sensitivity of model parameters to structural damping characteristic (Marra et al., 2015; Xu et al., 2020; Zhang et al., 2020). The difference in structural damping between the real bridge and the scaled section model from which the aerodynamic parameters are identified may cause unacceptable prediction error of VIV amplitude. This limitation can be overcome by using improved aerodynamic force models such as the describing function-based model (Mashnad and Jones, 2014; Zhang et al., 2019) and the generalized polynomial model (Xu et al., 2018), which directly simulate the amplitude-dependent damping effect of aerodynamic force. As there is no essential distinction between these two models, either one can be employed to derive a more reliable VIV analysis approach for bridges with multiple-cross-section deck, and the generalized polynomial model has been chosen in this study.

In this study, an optimization method is proposed for the spanwise layout of aerodynamic countermeasures to enhance the economic efficiency of VIV control. To accomplish the optimization, an improved mode-by-mode 3-D VIV analysis approach for bridges with multiple-cross-section deck is developed based on the generalized polynomial model. With constraints of limited multi-mode VIV responses, the genetic algorithm (GA) is used to search for the optimum spanwise layout which requires the fewest aerodynamic countermeasures. Two alternative types of search spaces are provided for GA. To verify the feasibility of the proposed method, a long-span suspension bridge with semi-closed

box deck is employed as a case study. Finite element simulation on the full bridge and wind tunnel tests on the bridge deck section model are conducted to obtain the structural and aerodynamic parameters required for the optimization. The aerodynamic countermeasure layout schemes optimized with different search spaces and additional constraints are compared. The impact of structural and aerodynamic parameters on the optimized result is discussed.

2. Methodology

2.1. 3-D VIV analysis approach for bridges with multiple-cross-section deck

The two potential forms of VIV are vertical and torsional vibrations, with the former being more prevalent in real bridges (Battista and Pfeil, 2000; Fujino and Yoshida, 2002; Ge et al., 2022; Larsen et al., 2000; Li et al., 2011; Macdonald et al., 2002). The motion equation governing the vertical VIV of a bridge deck section can be expressed as

$$m[\ddot{y}(t) + 2\omega_s \zeta_s \dot{y}(t) + \omega_s^2 y(t)] = F \quad (1)$$

where y is the vertical displacement; the over-dot symbol denotes differentiation with respect to time t ; m is the mass per unit length; ω_s and ζ_s are, respectively, the structural circular frequency and damping ratio; F is the vertical aerodynamic force per unit length.

To describe the 2-D aerodynamic force acting on bridge deck, the generalized polynomial model proposed by Xu et al. (2018) is adopted. This model employs a generalized polynomial function on structural vibration velocity to reproduce the nonlinear aerodynamic damping force, which is the primary component of the aerodynamic force that contributes to the amplitude of VIV response. The generalized polynomial model is expressed as

$$F = \rho U^2 D \left[\sum_i Y_i \frac{|\dot{y}(t)|^{i-1}}{U^{i-1}} \right] \frac{\dot{y}(t)}{U} \quad (2)$$

where ρ is the air density; U is the uniform wind speed; D is the depth of the deck section; Y_i is the aerodynamic parameter which relies on the reduced frequency $K = 2\pi f D / U$ (or reduced wind speed $U_r = U / f D$) and the aerodynamic shape of deck section; f is the vibration frequency. These aerodynamic parameters can be identified through wind tunnel tests on a bridge deck sectional model. The order of the generalized polynomial model applied on a specific deck section is determined according to the nonlinear characteristics of the aerodynamic damping force acting on this section.

Substituting Eq. (2) into Eq. (1) and rewriting the motion equation into a dimensionless form yield

$$m[\ddot{\eta}(s) + 2K_s \zeta_s \dot{\eta}(s) + K_s^2 \eta(s)] = \rho D^2 \left[\sum_i Y_i |\dot{\eta}(s)|^{i-1} \right] \dot{\eta}(s) \quad (3)$$

where $\eta = y/D$ is the normalized vertical displacement; the apostrophe denotes differentiation with respect to the dimensionless time $s = Ut/D$; $K_s = \omega_s D / U$ is the reduced structural frequency.

Since the aerodynamic parameters Y_i are related to the section shape, they could change with the location x along the span of the bridge deck, unless the cross-section is uniform. For example, the aerodynamic parameter of a bridge deck with two types of cross-sections (CsA and CsB) can be expressed as

$$Y_i(x) = \begin{cases} Y_{i,A}, & x \in X_A \\ Y_{i,B}, & x \in X_B \end{cases} \quad (4)$$

where $Y_{i,A}$ and X_A are, respectively, the aerodynamic parameter and the spanwise location set of CsA, while $Y_{i,B}$ and X_B are the counterparts for CsB. X_A and X_B satisfy

$$\mathbf{X}_A \cup \mathbf{X}_B = [0, L], \mathbf{X}_A \cap \mathbf{X}_B = \emptyset \quad (5)$$

where L is the total span of the bridge deck.

The aerodynamic parameters of each cross-section on the prototype bridge deck cannot maintain the 2-D values due to the influence of spanwise flow perturbation. Nevertheless, previous studies (Meng et al., 2015; Sun et al., 2019; Zhu et al., 2017b) demonstrate that this perturbation is relatively weak on the uniform bridge deck, and the self-excited force represented by Y_i can be regarded as fully correlated along the span. Although changing the shape of deck section may cause spanwise flow around the junction of two distinct cross-sections on multiple-cross-section deck, this effect tends to suppress vortex shedding and reduce structural VIV responses (El-Gammal et al., 2007; Xin et al., 2018; Zhang et al., 2022). Therefore, it is conservative to neglect this 3-D effect in bridge design. The aerodynamic parameter identification results of the 2-D sectional model can be directly substituted for $Y_{i,A}$ and $Y_{i,B}$ in Eq. (4). Moreover, reducing the number of spanwise junctions where the section shape changes can help to minimize the impact of neglecting such 3-D effect.

Based on in-situ observations (Battista and Pfeil, 2000; Fujino and Yoshida, 2002; Ge et al., 2022; Larsen et al., 2000; Li et al., 2011; Macdonald et al., 2002), it has been confirmed that the bridge oscillates in only one structural mode each time VIV occurs. Hence, the displacement response of the bridge deck can be expressed as

$$\eta(x, s) = \varphi(x)\nu(s) \quad (6)$$

where $\varphi(x)$ is the mode shape; $\nu(s)$ is the generalized coordinate.

By substituting Eq. (6) into Eq. (3), multiplying $\varphi(x)$ on both sides, and integrating over the entire span of the bridge deck, the governing equation for single-mode VIV analysis can be obtained as follows:

$$\tilde{M}[\nu''(s) + 2K_s\zeta_s\nu'(s) + K_s^2\nu(s)] = \rho D^2 \left[\sum_i \tilde{Y}_i |\nu'(s)|^{i-1} \right] \nu'(s) \quad (7)$$

where $\tilde{M}$ and $\tilde{Y}_i$ respectively denote the generalized mass and aerodynamic parameter:

$$\tilde{M} = \int_0^L m\varphi^2(x)dx \quad (8)$$

$$\tilde{Y}_i = \int_0^L Y_i(x)|\varphi(x)|^{i+1}dx \quad (9)$$

As $\varphi(x)$ in Eq. (8) only represents the mode shape of the bridge deck, m should be revised with the equivalent mass to consider the 3-D structural and vibrational behaviors of the bridge.

Solving Eq. (7) with numerical algorithm yields the displacement time history of the bridge deck. Since VIV is a kind of weakly nonlinear vibration, the steady-state amplitude can be obtained more easily using the equivalent linearization technique (Rice, 1995). The linearized representation of Eq. (7) is

$$\tilde{M}[\nu''(s) + 2K_s(\zeta_s + \zeta_a)\nu'(s) + K_s^2\nu(s)] = 0 \quad (10)$$

where ζ_a is the equivalent aerodynamic damping ratio. The expression of ζ_a can be determined from the following energy equivalence equation:

$$\int_0^{2\pi/K} -2\tilde{M}K_s\zeta_a\nu'(s) \cdot \nu'(s)ds = \int_0^{2\pi/K} \rho D^2 \left[\sum_i \tilde{Y}_i |\nu'(s)|^{i-1} \right] \nu'(s) \cdot \nu'(s)ds \quad (11)$$

Assuming a sinusoidal form for the generalized coordinate, i.e., $\nu(s) = A_\nu \sin(Ks)$, and substituting it into Eq. (11), yield

$$\zeta_a(A_\nu) = -\frac{\rho D^2}{2\tilde{M}K_s} \left[\sum_i \tilde{Y}_i \frac{2\Gamma\left(\frac{1}{2}(i+2)\right)}{\sqrt{\pi}\Gamma\left(\frac{1}{2}(i+3)\right)} K^{i-1} A_\nu^{i-1} \right] \quad (12)$$

where $\Gamma(\cdot)$ denotes the Gamma function.

The total damping ratio ζ of the structural mode can thus be written as

$$\zeta(A_\nu) = \zeta_s + \zeta_a(A_\nu) = \zeta_s - \frac{\rho D^2}{2\tilde{M}K_s} \left[\sum_i \tilde{Y}_i \frac{2\Gamma\left(\frac{1}{2}(i+2)\right)}{\sqrt{\pi}\Gamma\left(\frac{1}{2}(i+3)\right)} K^{i-1} A_\nu^{i-1} \right] \quad (13)$$

In Eq. (13), the structural damping ratio and the first term ($i = 1$) of the aerodynamic damping ratio are independent of vibration amplitude A_ν and thereby represent the modal damping characteristic at amplitudes close to zero. The sum of them can serve as an evaluation index for VIV occurrence and is denoted as ζ_0 :

$$\zeta_0 = \zeta_s - \frac{\rho D^2}{2\tilde{M}K_s} \tilde{Y}_1 \quad (14)$$

If $\zeta_0 \geq 0$, the structural response initialized by small perturbation will decay to zero or maintain the vibration with negligible amplitude, indicating the absence of VIV. Otherwise, the vibration amplitude will grow until the total damping ratio equals zero, i.e., $\zeta(A_\nu) = 0$, and the stable VIV amplitude is thus determined.

2.2. Aerodynamic countermeasure spanwise layout optimization for single-mode VIV control

In the present study, the spanwise layout of aerodynamic countermeasures on the bridge deck is to be optimized with constraints of limited VIV responses. In this optimization, the entire bridge deck is composed of two types of cross-sections: the original VIV-prone section (CsA) and the modified section with aerodynamic countermeasures (CsB). The spanwise locations of these two cross-sections are respectively defined as $\mathbf{X}_A$ and $\mathbf{X}_B$, which also represents the arrangement of aerodynamic countermeasures along the span. According to Eq. (5), $\mathbf{X}_A$ is uniquely determined by $\mathbf{X}_B$. Therefore, $\mathbf{X}_B$ can be taken as the only design variable, and the optimization object is to minimize the total length of $\mathbf{X}_B$, i.e., the total installation length of aerodynamic countermeasures along the span.

Since VIV will more or less affect the bridge operation, it is preferable to avoid this type of vibration. As explained in the preceding subsection, the occurrence of VIV in a given mode can be assessed by the index ζ_0 defined by Eq. (14). It should be noted that the value of ζ_0 varies with reduced wind speed and initial attack angle α_0 . This is due to the dependence of the generalized aerodynamic parameter $\tilde{Y}_1$ on the reduced wind speed and aerodynamic shape of bridge deck section, and the latter will change at different initial attack angles. Accordingly, to fully control the VIV for a specific mode within the design ranges of attack angle ($\alpha_{0,d}$) and reduced wind speed ($U_{r,d}$), the following equation should be satisfied:

$$\zeta_{0,\min} = \min_{\alpha_0 \in \alpha_{0,d}, U_r \in U_{r,d}} \zeta_0(\alpha_0, U_r) \geq 0 \quad (15)$$

where $\zeta_{0,\min}$ is the minimum value of ζ_0 within the design ranges of attack angle and reduced wind speed.

Given the requirement to control the VIV for a single structural mode, the optimization problem can be formulated as

$$\min \ell(\mathbf{X}_B), \text{ s.t. } \zeta_{0,\min} \geq 0 \quad (16)$$

where $\ell(\cdot)$ denotes the total length of the location set.

Substituting Eqs. (4) and (9) into Eq. (14) yields the relation between

ζ_0 and $\mathbf{X}_B$:

$$\zeta_0(\alpha_0, U_r) = \zeta_s - \frac{\rho D^2}{2MK_s} \left[\int_{x \in \mathbf{X}_A} Y_{1,A}(\alpha_0, U_r) \varphi^2(x) dx + \int_{x \in \mathbf{X}_B} Y_{1,B}(\alpha_0, U_r) \varphi^2(x) dx \right] \quad (17)$$

where $Y_{1,A}$ and $Y_{1,B}$ are the aerodynamic parameters of CsA and CsB, respectively.

To facilitate the optimization, Eq. (17) is rewritten as

$$\zeta_0(\alpha_0, U_r) = \zeta_s - \frac{\rho D^2}{2MK_s} [(1-p)Y_{1,A}(\alpha_0, U_r) + pY_{1,B}(\alpha_0, U_r)] \int_0^L \varphi^2(x) dx \quad (18)$$

where

$$p = \frac{\int_{x \in \mathbf{X}_B} \varphi^2(x) dx}{\int_0^L \varphi^2(x) dx} \quad (19)$$

The introduced dimensionless parameter p ($0 \leq p \leq 1$) is closely related to the design variable $\mathbf{X}_B$. As shown in Eq. (18), p is the weight coefficient of $Y_{1,B}$, thus representing the contribution of installed aerodynamic countermeasures to VIV control. The derivative dp/dx quantifies the contribution of the aerodynamic countermeasure per unit length, which is proportional to $\varphi^2(x)$ of the countermeasure installation location. Therefore, to achieve the optimum control efficiency and minimize the use of aerodynamic countermeasures, these countermeasures should be preferentially installed at the spanwise locations with larger absolute values of mode shape, i.e., location x_k has priority over x_{k+1} if $|\varphi(x_k)| > |\varphi(x_{k+1})|$. Such an installation sequence leads to a specific relationship between $\mathbf{X}_B$ and $\ell(\mathbf{X}_B)$, subsequently determining the variation of p with $\ell(\mathbf{X}_B)$. Since p monotonically increases with the increase of $\ell(\mathbf{X}_B)$, the optimization problem described in Eq. (16) can be equivalently transformed to identify the minimum p -value that satisfies $\zeta_{0,\min} \geq 0$, along with the corresponding $\mathbf{X}_B$.

The relation between p and $\zeta_{0,\min}$ can be inferred as follows: $p = 0$ indicates that no aerodynamic countermeasure is installed on the bridge deck, and VIV will occur in some cases, leading to $\zeta_{0,\min} < 0$; $p = 1$ indicates that the entire bridge deck adopts CsB section with aerodynamic countermeasures, and it has been experimentally confirmed that no VIV occurs within the whole design ranges of attack angle and reduced wind speed, resulting in $\zeta_{0,\min} > 0$; Accordingly, there must exist a critical value p_{cr} between 0 and 1 that satisfies $\zeta_{0,\min} = 0$, which is the critical condition to prevent VIV in this mode. Because p_{cr} is also the minimum value of p satisfying $\zeta_{0,\min} \geq 0$, the corresponding $\ell(\mathbf{X}_B)$ and $\mathbf{X}_B$ are respectively the optimized installation length and location set of aerodynamic countermeasures. Based on Eq. (18), p_{cr} can be deduced from the following equation:

$$\min \left\{ \zeta_s - \frac{\rho D^2}{2MK_s} [(1-p_{cr})Y_{1,A}(\alpha_0, U_r) + p_{cr}Y_{1,B}(\alpha_0, U_r)] \int_0^L \varphi^2(x) dx \right\} = 0, \alpha_0 \in \alpha_{0,d}, U_r \in U_{r,d} \quad (20)$$

The optimization process for single-mode VIV control can be summarized as follows:

- (1) Prioritize the spanwise locations according to the mode shape. The aerodynamic countermeasures are preferentially installed at the locations with larger absolute values of mode shape;
- (2) Determine the variations of $\mathbf{X}_B$ and p with respect to $\ell(\mathbf{X}_B)$;
- (3) Calculate the critical value p_{cr} using Eq. (20);
- (4) Obtain the optimized countermeasure installation length $\ell(\mathbf{X}_B)$ and location set $\mathbf{X}_B$ corresponding to p_{cr} .

2.3. Aerodynamic countermeasure spanwise layout optimization for multi-mode VIV control

As long-span bridges are flexible structures with low and crowded modal frequencies, the occurrence of VIV can be prevalent across multiple structural modes within the design wind speed range. To effectively control VIV in all these modes, the spanwise layout optimization for aerodynamic countermeasures should be subject to the following equation:

$$\min_{\alpha_0 \in \alpha_{0,d}, U_r \in U_{r,d}} \zeta_{0,j}(\alpha_0, U_r) \geq 0, j \in \{1, 2, \dots, J\} \quad (21)$$

where $\zeta_{0,j}$ is the damping ratio of the j th to-be-controlled mode at amplitudes close to zero; J is the total number of target modes.

According to the preceding subsection, Eq. (21) is equivalent to the following equation:

$$p_j \geq p_{cr,j}, j \in \{1, 2, \dots, J\} \quad (22)$$

where p_j and $p_{cr,j}$ are the VIV control parameter and corresponding critical value of the j th mode.

Due to the significant difference in vibration shape among the target modes, it is challenging to prioritize the installation locations for aerodynamic countermeasures in advance, which complicates the optimization problem for multi-mode VIV control. Accordingly, the genetic algorithm (GA) is employed to get the global optimum solution. GA simulates the evolution of the population that is a collection of candidate solutions (i.e., individuals of GA). This algorithm consists of three key steps: (1) coding of design variables into string-type individuals; (2) evaluation of individuals based on objective function values; (3) application of genetic operators such as selection, crossover, mutation and reinsertion, to produce next generation of individuals.

The design variable considered in this study is the spanwise layout of aerodynamic countermeasures. In practical applications, these countermeasures are usually installed in discrete segments along the bridge deck. Therefore, the entire bridge deck can be divided into several spanwise segments for countermeasure installation, and the optimization aims to find the optimum solution from finite countermeasure layout schemes across these bridge deck segments. However, direct coding of countermeasure layout will yield a large number of individuals that do not satisfy the constraints presented in Eq. (21) or Eq. (22), thus affecting the optimization efficiency of GA. To transform the constrained optimization problem into an equivalent unconstrained problem, the individual of GA is defined as a prioritization of the bridge deck segments for the installation of aerodynamic countermeasures:

$$\mathbf{I} = \{\delta_1, \delta_2, \dots, \delta_n, \dots, \delta_N\} \quad (23)$$

where $\{\delta_1, \delta_2, \dots, \delta_N\}$ is a permutation of $\{1, 2, \dots, N\}$; $\{1, 2, \dots, N\}$ is the numbering set of all bridge deck segments. n is the minimum value that satisfies the following equation:

$$p_j = \frac{\int_{x \in \cup_{k=1}^n \mathbf{X}_{\delta_k}} \varphi_j^2(x) dx}{\int_0^L \varphi_j^2(x) dx} \geq p_{cr,j}, j \in \{1, 2, \dots, J\} \quad (24)$$

where $\mathbf{X}_{\delta_k}$ is the spanwise location set of the δ_k th bridge deck segment; $\varphi_j(x)$ is the vibration shape of the j th mode. Eq. (24) is an implicit constraint for the optimization.

According to Eqs. (23) and (24), $\ell(\cup_{k=1}^n \mathbf{X}_{\delta_k})$ is the minimum required

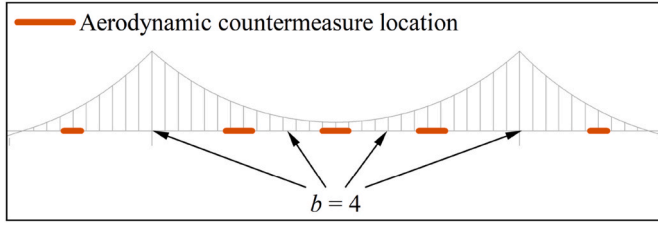


Fig. 1. Schematic diagram of the breaks among the aerodynamic countermeasure installation locations.

installation length of aerodynamic countermeasures to fully control all VIV-prone modes under current prioritization of bridge deck segments, which is taken as the primary objective function (Q_1) to be minimized in the optimization. The union set $\cup_{k=1}^n X_{\delta_k}$ is the optimized installation location range of aerodynamic countermeasures. However, owing to the remarkable difference in vibration shape among all target modes, the optimized countermeasure layout could be excessively scattered along the span, which will compromise the reliability of the proposed 3-D VIV

analysis approach and adversely affect the aesthetic appeal of the bridge. Therefore, a secondary objective function (Q_2) related to the number of breaks in $\cup_{k=1}^n X_{\delta_k}$ is introduced to make the optimized installation location set as continuous as possible. The objective function of GA is thus defined as

$$Q = Q_1 + Q_2 = \mathcal{L}(\cup_{k=1}^n X_{\delta_k}) + \frac{b}{n} \min_{k=1 \sim N} \mathcal{L}(X_k) \quad (25)$$

where b is the number of breaks in $\cup_{k=1}^n X_{\delta_k}$, which is illustrated in Fig. 1. Q_1 has priority over Q_2 in the optimization because $0 \leq b/n < 1$. As a result, the minimum required installation length of aerodynamic countermeasures for multi-mode VIV control can be determined from the optimization, and the countermeasure spanwise layout with optimum continuity can be further identified from the individuals with the minimum value of Q_1 .

With the individual defined as Eq. (23), all potential permutations of $\{1, 2, \dots, N\}$ constitute the global search space for the optimization, and the size of this search space will rapidly increase with the total number N of divided bridge deck segments. However, some segments, such as those located near the bridge bearings, are unlikely to be selected for the

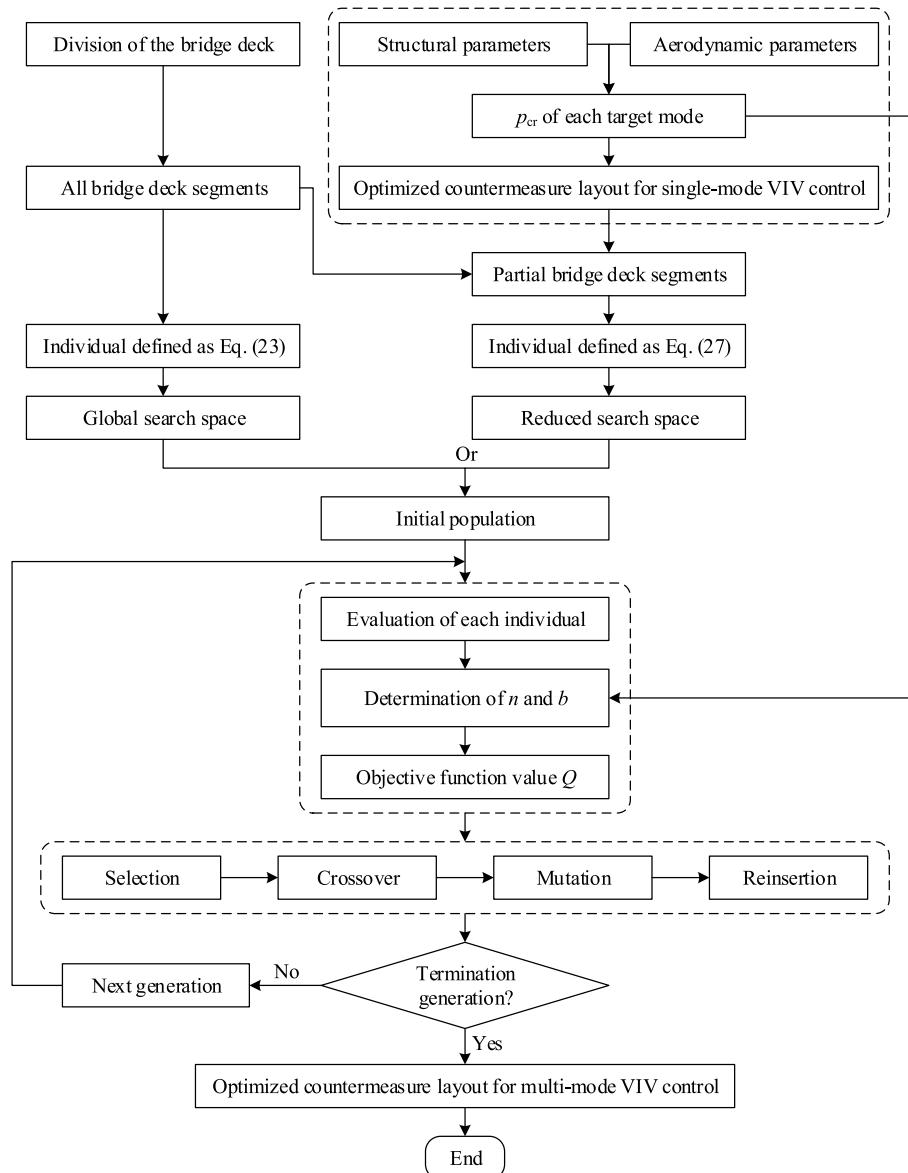


Fig. 2. Flow chart of the optimization.

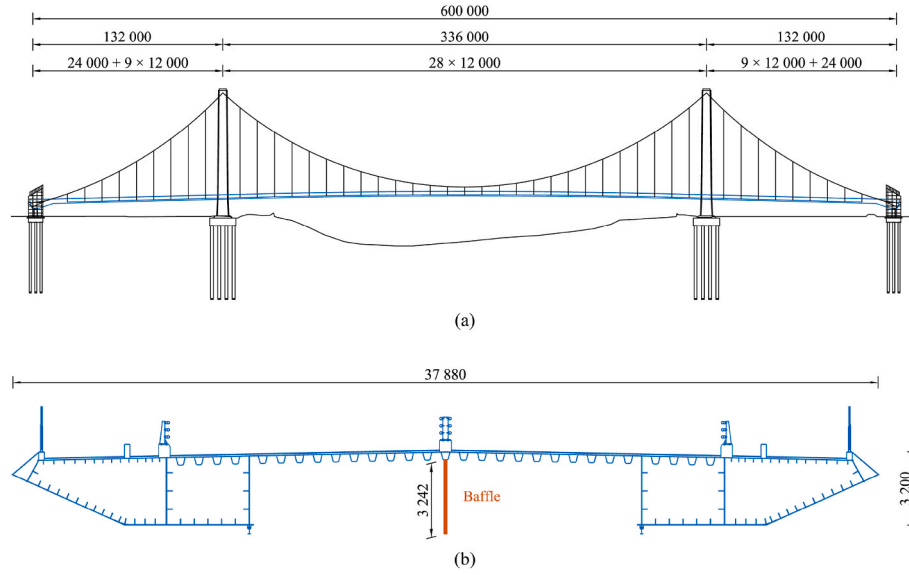


Fig. 3. Layout of the suspension bridge (unit: mm): (a) elevation; (b) cross-section of the bridge deck. The cross-sections without and with the baffle are respectively defined as CsA and CsB.

installation of aerodynamic countermeasures in the optimized layout scheme. Prior removal of these segments from $\{1, 2, \dots, N\}$ can help to reduce the search space and thus improve the search efficiency for optimum solution. As proposed in the previous subsection, the single-mode optimization method can be used to identify the optimum countermeasure installation locations for VIV control in each target mode, i.e., $\mathbf{X}_{B,j}$ ($j = 1-J$). Given that these locations are also with high priority for multi-mode VIV control, the union of them, i.e., $\cup_{j=1}^J \mathbf{X}_{B,j}$, can be utilized to restrict the installation location range of aerodynamic countermeasures and reduce the search space in the optimization. Accordingly, only partial bridge deck segments are allowed to install countermeasures, which can be extracted through the following equation:

$$\mathbf{S} = \left\{ k \in \{1, 2, \dots, N\} \mid \mathbf{X}_k \cap \left(\cup_{j=1}^J \mathbf{X}_{B,j} \right) \neq \emptyset \right\} \quad (26)$$

where $\mathbf{S}$ is the numbering set of the extracted bridge deck segments. The prioritization of these segments forms another definition of the GA individual:

$$\mathbf{I} = \{\delta_1, \delta_2, \dots, \delta_n, \dots, \delta_s\} \quad (27)$$

where s is the total number of the selected bridge deck segments in $\mathbf{S}$; $\{\delta_1, \delta_2, \dots, \delta_s\}$ is a permutation of $\mathbf{S}$; n is the minimum value that satisfies Eq. (24). All potential permutations of $\mathbf{S}$ constitute the reduced search space. The objective function of Eq. (27) is also defined as Eq. (25). Since $\cup_{k=1}^s \mathbf{X}_{\delta_k}$ is an over-conservative countermeasure layout scheme satisfying $p_j > p_{cr,j}$ ($j = 1-J$), there must exist an optimum solution $\cup_{k=1}^n \mathbf{X}_{\delta_k}$ with smaller countermeasure installation length.

Fig. 2 shows the flow chart of the optimization with global or reduced search space. Optimization conducted within the global search space yields the global optimum spanwise layout of aerodynamic countermeasures, while optimization within the reduce search space lead to a suboptimal solution. Nevertheless, the total countermeasure installation length of this suboptimal solution is small enough, and further evidence will be given in section 4.2.

3. Application example

3.1. Structural characteristics of the prototype bridge

A self-anchored suspension bridge is chosen as the case study to evaluate the effectiveness of the proposed method. As shown in Fig. 3,

Table 1

Structural dynamic characteristics of the suspension bridge.

Mode No.	Frequency (Hz)	Equivalent mass of the bridge deck (kg/m)	Mode shape
1	0.3029	37998	
2	0.3188	38254	
3	0.5421	37445	
4	0.5571	46106	

the bridge has a total span of 600 m, and the deck adopts a semi-closed box type with dimensions of 37.88 m width and 3.2 m depth. An additional baffle is installed at the centre of the lower slot of bridge deck as the aerodynamic countermeasure for VIV, and its effectiveness is experimentally verified and described in the next subsection. The bridge deck cross-sections without and with this baffle are respectively defined as CsA and CsB in this study. A finite element model of the bridge was established to analyse the structural dynamic characteristics. The results of the first 4 vertical modes, which are prone to VIV within the design wind speed range 0–25 m/s, are listed in Table 1, and all of them should be controlled. The damping ratio of each mode is assumed as 0.3%.

3.2. Aerodynamic parameters of the bridge deck sections

Wind tunnel tests on the bridge deck section model were carried out to validate the effectiveness of the baffle and identify the aerodynamic parameters required for the spanwise layout optimization of baffles on the bridge. The bridge deck sections including CsA and CsB were simulated using a sectional model with a length scale of 1:50. The size of the model is 1.740 m (length) $\times$ 0.732 m (width) $\times$ 0.064 m (depth). The VIV tests on the sectional model were conducted in the TJ-1 wind tunnel at Tongji University. The test section of the wind tunnel is 1.8 m in width and 1.8 m in height. As shown in Fig. 4, the sectional model was suspended by eight helical springs through two arms, and the vibration responses of the model were recorded by the laser displacement sensors. To ensure a 2-D flow field around the sectional model, the supporting and measuring devices were placed outside the wind tunnel and the elliptical plates were attached to both ends of the model.

The mass and natural frequency of the sectional model are 26.514 kg

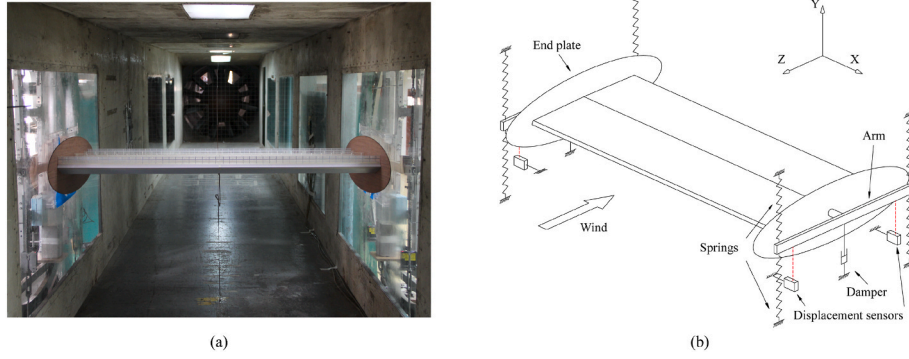


Fig. 4. System of the spring-suspended sectional model: (a) mounted model in TJ-1 wind tunnel; (b) schematic diagram of the experimental setup.

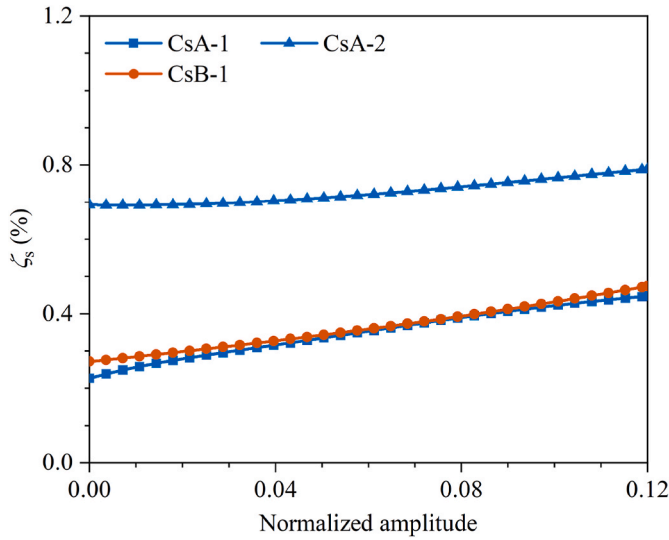


Fig. 5. Structural damping ratios of the sectional model.

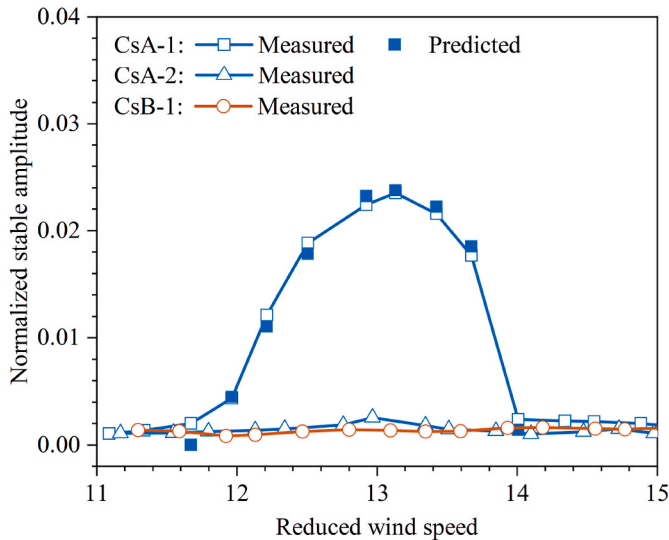


Fig. 6. Measured and predicted vertical steady-state amplitudes of the sectional model.

and 3.735 Hz. External dampers, as shown in Fig. 4(b), were applied to adjust the structural damping. Two types of damping ratios were involved in the test: a low damping ratio of approximately 0.3% (close to

the design damping ratio) and a high damping ratio of approximately 0.7%. The low damping was adopted in test cases CsA-1 and CsB-1, which were designed to verify the effectiveness of the baffle and identify the aerodynamic parameters of the modified section CsB. The high damping was adopted in test case CsA-2, which was mainly used to identify the aerodynamic parameters of the original section CsA. As there inevitably exist nonlinearities in damping behaviors of the sectional model, the structural damping ratio will vary with vibration amplitude rather than being constant. The time domain method presented by Gao and Zhu (2015) was employed to identify the amplitude-dependent damping ratio of the sectional model, and the results are shown in Fig. 5.

The tests were carried out under smooth wind flow with turbulence intensity lower than 1%. The VIV-prone attack angle of $+3^\circ$ is taken as an example in this study, and the observed vertical steady-state amplitudes of the sectional model are shown in Fig. 6. Significant VIV occurs on the original bridge deck section CsA in case CsA-1. The lock-in range of reduced wind speed is from 11.67 to 14.01. Comparison between cases CsA-1 and CsB-1 shows that the VIV is fully suppressed with application of the baffle.

As suggested by Xu et al. (2018), the aerodynamic parameters in Eq. (2) can be determined in two steps: (1) identify the aerodynamic damping ratio of the sectional model; (2) fit the identified aerodynamic damping ratio and aerodynamic parameters. Decay-to-resonance tests were conducted in cases CsA-2 and CsB-1, wherein the VIV is fully suppressed, to extract the aerodynamic damping ratios of CsA and CsB. To be specific, the sectional model was artificially excited to oscillate and then freely decayed to zero, as shown in Fig. 7. With the measured displacement time histories, the total amplitude-dependent damping ratio of the aeroelastic system was identified with Gao and Zhu's method (Gao and Zhu, 2015). The aerodynamic damping ratio was then obtained by subtracting the structural damping ratio from the total one. By repeating the aforementioned identification process during the lock-in range, the variation of aerodynamic damping ratio with the vibration amplitude and reduced wind speed was obtained, which are shown as the experimental results in Fig. 8.

Similar to Eq. (12), the relation between the aerodynamic damping ratio and aerodynamic parameters of the sectional model can be expressed as

$$\zeta_a(A_\eta) = -\frac{\rho D^2}{2mK_s} \left[\sum_i Y_i \frac{2\Gamma\left(\frac{1}{2}(i+2)\right)}{\sqrt{\pi}\Gamma\left(\frac{1}{2}(i+3)\right)} K^{i-1} A_\eta^{i-1} \right] \quad (28)$$

where A_η is amplitude of the normalized displacement η . With the initial attack angle set as $+3^\circ$, the aerodynamic parameter of each section is only dependent on the reduced wind speed, and their relation can be approximated by a polynomial function:

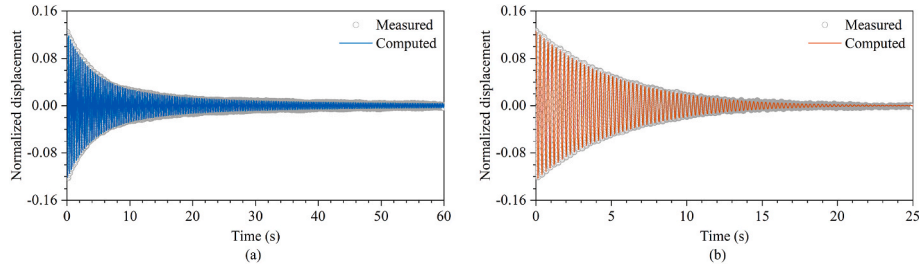


Fig. 7. Time history comparison between the measured and computed displacements. (a) case CsA-2, $U_r = 13.34$; (b) case CsB-1, $U_r = 13.34$.

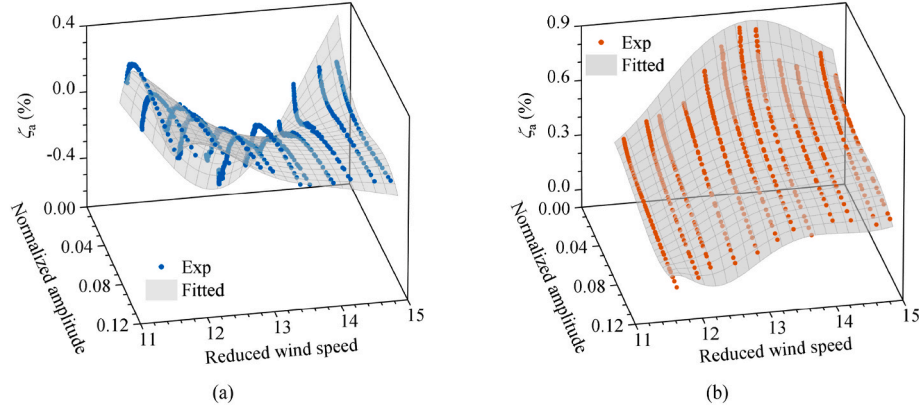


Fig. 8. Aerodynamic damping ratios of the sectional model: (a) CsA; (b) CsB.

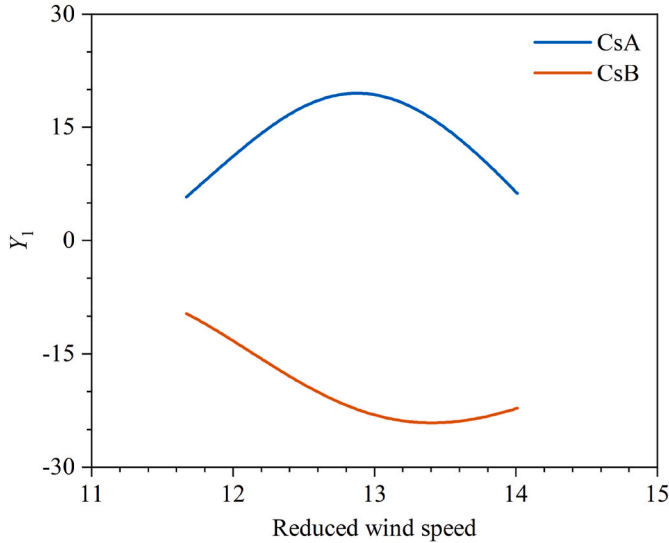


Fig. 9. Identified aerodynamic parameter Y_1 .

$$Y_i(U_r) = \sum_k q_{ik} U_r^{k-1} \quad (29)$$

where q_{ik} is to-be-determined parameter.

Substituting Eq. (29) into Eq. (28) yields

$$\zeta_a(U_r, A_\eta, q_{ik}) = -\frac{\rho D^2}{2mK_s} \left[\sum_i \sum_k q_{ik} \frac{2\Gamma\left(\frac{1}{2}(i+2)\right)}{\sqrt{\pi}\Gamma\left(\frac{1}{2}(i+3)\right)} U_r^{k-1} \left(\frac{2\pi}{U_r}\right)^{i-1} A_\eta^{i-1} \right] \quad (30)$$

Based on Eq. (30), q_{ik} can be identified through least squares fitting on the experimental aerodynamic damping ratio, wherein the residual is

defined as

$$R(q_{ik}) = \sum_{z=1}^Z [\zeta_a(\hat{U}_{r,z}, \hat{A}_{\eta,z}, q_{ik}) - \hat{\zeta}_{a,z}]^2 \quad (31)$$

where the superscript $\hat{\cdot}$ represents the data acquired through experiments; $(\hat{U}_{r,z}, \hat{A}_{\eta,z}, \hat{\zeta}_{a,z})$ is corresponding to a single experimental data point shown in Fig. 8; Z is the total number of experimental data points. It was found that satisfactory fitting results can be achieved by taking $i = 1-4$ and $k = 1-7$ in Eq. (30), and the fitted aerodynamic damping ratios are shown in Fig. 8. With q_{ik} determined, the aerodynamic parameters Y_i ($i = 1-4$) can be obtained from Eq. (29).

As a verification, the identified aerodynamic parameters are used to numerically compute the displacements of the sectional model, which agree well with the experimental results in Fig. 7. The aerodynamic parameters obtained from case CsA-2 are also used to predict the steady-state VIV amplitudes of case CsA-1. Fig. 6 demonstrates a close alignment between the predicted and measured amplitudes. This cross validation shows the insensitivity of the aerodynamic parameters to structural dynamic characteristics, as well as the reliability of the applied aerodynamic force model to predict VIV amplitudes.

Fig. 9 shows the identified aerodynamic parameters (Y_1) of CsA and CsB, which will be used in the subsequent optimization. Y_1 of CsA keeps positive within the lock-in range, indicating that the corresponding linear aerodynamic damping force acting on CsA is negative and thus induces vibration. Conversely, Y_1 of CsB becomes negative and leads to positive aerodynamic damping, which suggests that the baffle has strong control effect for VIV.

4. Results and discussion

4.1. Optimized layout schemes of aerodynamic countermeasures for single-mode VIV control

To control the VIV in only one mode of the bridge at $+3^\circ$ attack

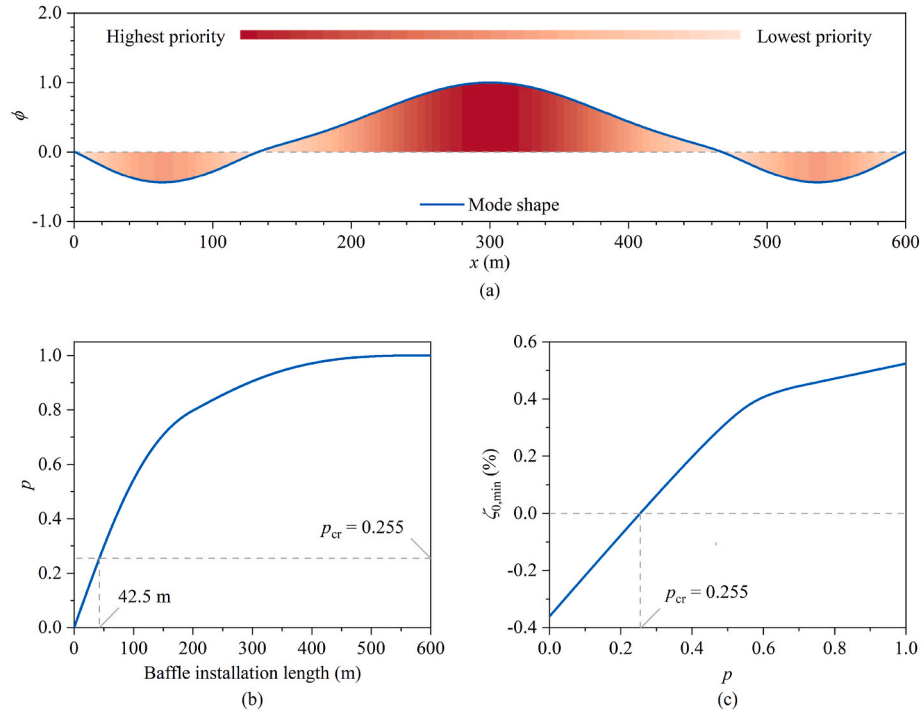


Fig. 10. Spanwise layout optimization process for aerodynamic countermeasures to control the VIV in Mode 1: (a) priority of countermeasure installation locations; (b) variation of p with countermeasure installation length; (c) variation of $\zeta_{0,\min}$ with p .

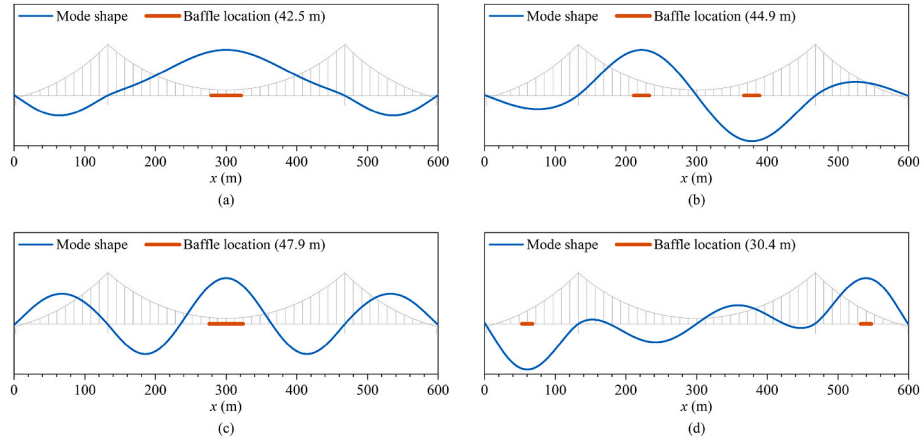


Fig. 11. Optimized spanwise layout schemes of aerodynamic countermeasures for single-mode VIV control: (a) Mode 1; (b) Mode 2; (c) Mode 3; (d) Mode 4.

angle, the spanwise layout of the applied aerodynamic countermeasure (i.e., the baffle shown in Fig. 3(b)) can be optimized using the proposed single-mode optimization method in conjunction with the identified structural and aerodynamic parameters. Taking Mode 1 as an example, the optimization process is illustrated in Fig. 10. Fig. 10(a) shows the prioritization of the spanwise locations to install baffles. The location with larger absolute value of mode shape is given higher priority. With such an installation sequence, the variation of p , defined as Eq. (19), with baffle installation length is determined as shown in Fig. 10(b). The curve slope represents the contribution of the baffle per unit length to VIV control, which monotonously decreases to zero with the increase of total baffle installation length. Therefore, the defined baffle installation sequence provides the optimum VIV control effect with identical baffle installation length compared to other schemes. Fig. 10(c) shows the variation of $\zeta_{0,\min}$ with p , where $\zeta_{0,\min}$ is the minimum value of $\zeta_0(+3^\circ, U_r)$ within the lock-in range of wind speed. The VIV control effect represented by $\zeta_{0,\min}$ is improved with the increase of p . $\zeta_{0,\min} = 0$ is the

critical condition to prevent VIV, which can be achieved by a threshold p -value, i.e., $p_{cr} = 0.255$. Consequently, $p \geq p_{cr}$ indicates reliable control for VIV, and p_{cr} is the minimum required value. The minimum required baffle installation length (42.5 m) for VIV control of Mode 1 can thereby be determined from $p = p_{cr}$ in Fig. 10(b), and the corresponding location range is the optimized spanwise layout scheme for baffles.

Fig. 11 shows the optimized baffle layout schemes for Modes 1–4. These layout schemes differ in length and location due to the variation of generalized mass and vibration shape associated with each target mode. Taking Mode 1 as an example in Fig. 12, the control effect of the optimized baffle layout scheme is compared with those of other layout schemes that either have no baffle or have baffles installed along the entire span in terms of ζ_0 -value. When no baffle is installed on the bridge deck, VIV occurs at the reduced wind speed that satisfies $\zeta_0(+3^\circ, U_r) < 0$. When baffles are installed along the entire span, $\zeta_{0,\min} > 0$ within the entire reduced wind speed range, indicating complete suppression of VIV but an overly conservative control effect. In contrast, the optimized

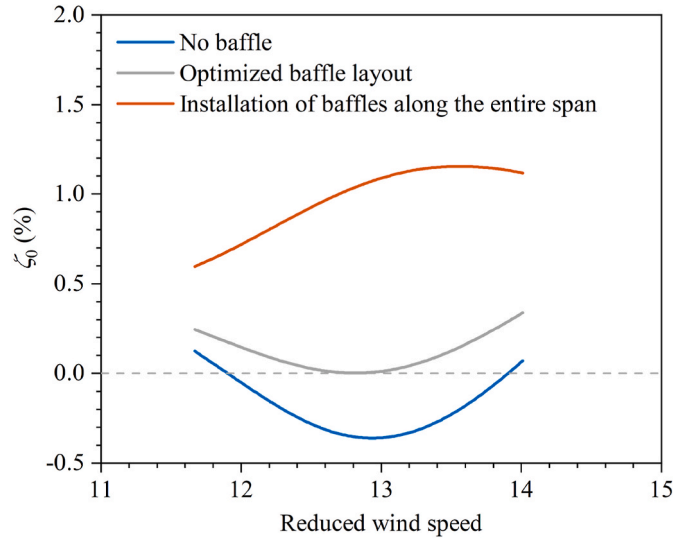


Fig. 12. ζ_0 of Mode 1 under different aerodynamic countermeasure layout schemes.

baffle layout scheme shown in Fig. 11(a) achieves the critical control state to prevent the VIV of Mode 1, which satisfies $\zeta_{0,\min} = 0$.

4.2. Optimized layout schemes of aerodynamic countermeasures for multi-mode VIV control

To simultaneously control the 4 VIV-prone modes of the bridge, the spanwise layout optimization of baffles should be carried out using the GA. Since the baffles need to be installed by connecting their ends to the bridge deck diaphragms, the basic bridge deck segment for the optimization is set to be the segment between two diaphragms, with a spanwise length of 4 m. Two types of search spaces are employed in GA: a global search space with the individual defined as Eq. (23), and a reduced search space with the individual defined as Eq. (27) based on the single-mode optimization results shown in Fig. 11. The allowable baffle

installation locations of the global and reduced search space are respectively shown in Fig. 13(a) and (b). Furthermore, the bridge shown in Fig. 3(a) is symmetric along the span, which is a common situation in engineering practice. Therefore, it is worthwhile to search for the symmetric optimized layout, which can be realized by considering only the first half of the bridge deck segments in Eq. (23) or Eq. (27) and multiplying p_j by 2 in Eq. (24). This operation is referred as additional symmetry constraint for ease of discussion. To compare the results of the optimization with different search spaces and additional symmetry constraints, 4 optimization cases are designed and listed in Table 2. The control parameters of GA for each case are provided in Table 3.

Fig. 13(a) and (b) show the optimized baffle layout schemes of Cases 1–4, wherein the total installation length of baffles is notably reduced in each layout scheme. Cases 1 and 3, without additional symmetry constraints, yield optimized layout schemes that are asymmetric along the span, although the bridge is symmetric. The optimized symmetric layout schemes of Cases 2 and 4 requires slightly more baffles than the asymmetric results of Cases 1 and 3, but the symmetric layout is superior in terms of construction convenience and aesthetics. In the optimized

Table 2
Optimization cases of GA.

Case No.	Search space	Additional symmetry constraint
1	Global search space	No
2	Global search space	Yes
3	Reduced search space	No
4	Reduced search space	Yes

Table 3
Control parameters of GA.

Parameter	Value
Population size	50
Generation gap	0.95
Termination generation	2000
Crossover probability	0.8
Mutation probability	0.05

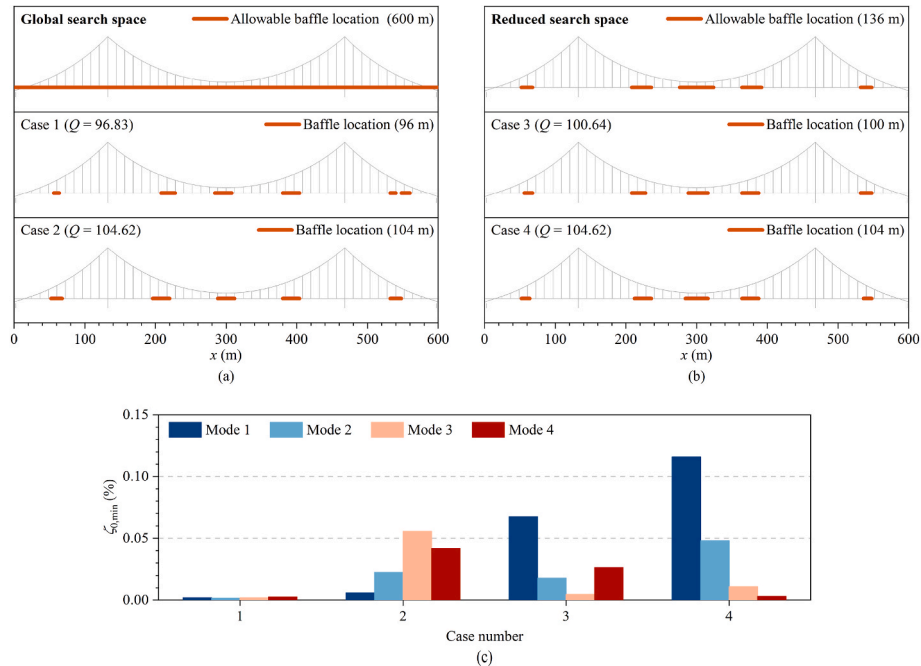


Fig. 13. Optimized spanwise layout schemes of aerodynamic countermeasures for multi-mode VIV control: (a) optimized results from the global search space; (b) optimized results from the reduced search space; (c) $\zeta_{0,\min}$ of the target modes under different optimized layout schemes.

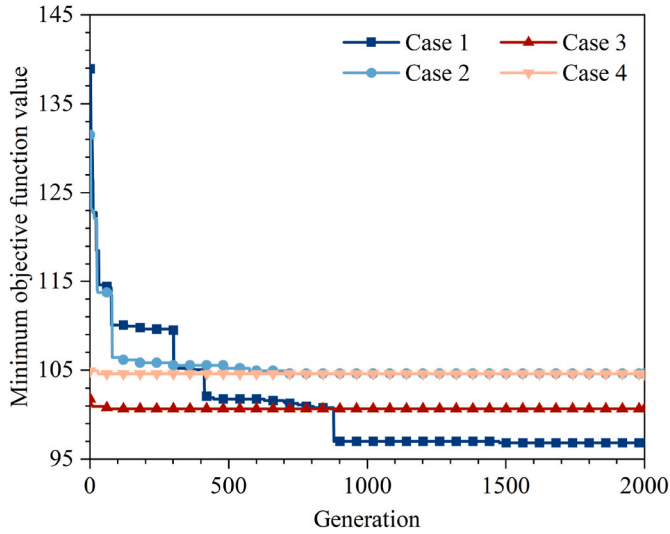


Fig. 14. Convergence of objective function in different optimization cases.

layout schemes of Cases 1 and 2, the chosen installation locations for baffles fall within or in close proximity to the allowable location range of the reduced search space. Therefore, although the baffle installation locations are limited to a small range, Cases 3 and 4 can achieve layout schemes with very similar or even identical lengths of baffles compared to the results of Cases 1 and 2.

In Fig. 13(c), $\zeta_{0,\min}$ of the target mode is used to describe and compare the control effects of the optimized baffle layout schemes for VIV. The results show that all layout schemes effectively prevent VIV in the target modes, as $\zeta_{0,\min} > 0$. For the optimized layout scheme of Case 1, $\zeta_{0,\min}$ of each target mode only slightly exceeds zero, indicating that this scheme provides a comprehensive control effect for multi-mode VIV that is very close to the critical state. This is because Case 1 has the fewest constraints in the optimization. If the bridge deck is divided into smaller segments for GA, the control effect of the optimized layout scheme will be closer to the critical state. The increased installation

lengths of baffles in the optimized layout schemes of Cases 2–4 lead to more conservative control for VIV in some modes, but there is at least one mode wherein $\zeta_{0,\min}$ approaches zero.

Fig. 14 compares the optimization history of Cases 1–4 in terms of the minimum objective function value among the individuals. The convergence of Cases 3 and 4 is achieved within 70 generations, which are much fewer than that required for Cases 1 and 2. This is because the size of the reduced search space is much smaller than the global space. Hence, conducting the optimization with the reduced search space is more practical and efficient.

It should be emphasized that the optimum solution obtained using the proposed method may not be unique, which is because the baffles are installed in discrete segments along the bridge deck. Taking Case 2 as an example, the non-uniqueness of the optimum solution is demonstrated by comparing the results of repeated optimization. As shown in Fig. 15(a), the optimized Layouts 1–3 have the same objective function value but different baffle layouts. Nevertheless, it can be seen from Fig. 15(c) that $\zeta_{0,\min}$ approaches zero for at least one mode in each layout scheme, indicating little difference in the VIV control effect among Layouts 1–3. Accordingly, any of these optimized layout schemes can be chosen for practical application.

The necessity of introducing Q_2 in Eq. (25) is also discussed here. Layouts 4–6 shown in Fig. 15(b) are obtained from the optimization without Q_2 . Although the required baffle installation length of Layouts 4–6 is equal to that of Layouts 1–3, the locations of baffles in Layouts 4–6 are excessively scattered along the span. Consequently, the flow field around the bridge deck becomes much more complex due to the perturbation from 3-D spanwise flow, which compromises the reliability of the computed structural response using the proposed 3-D VIV analysis approach. Moreover, the appearance of the bridge is notably affected by the scattered layout of baffles.

4.3. Impact of structural and aerodynamic parameters on the optimized results

Due to the dependence of the proposed optimization method on the relative relation between structural and aerodynamic damping ratios, variations in the input structural and aerodynamic damping parameters

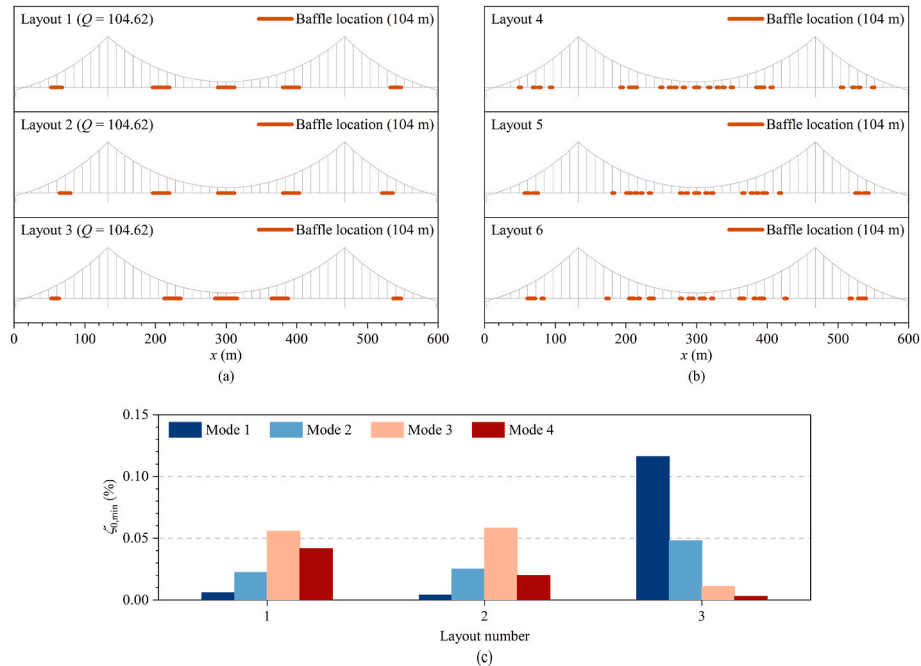


Fig. 15. Aerodynamic countermeasure layout schemes obtained from repeated optimization with the objective function defined as (a) $Q_1 + Q_2$ or (b) Q_1 ; (c) $\zeta_{0,\min}$ of the target modes under different layout schemes.

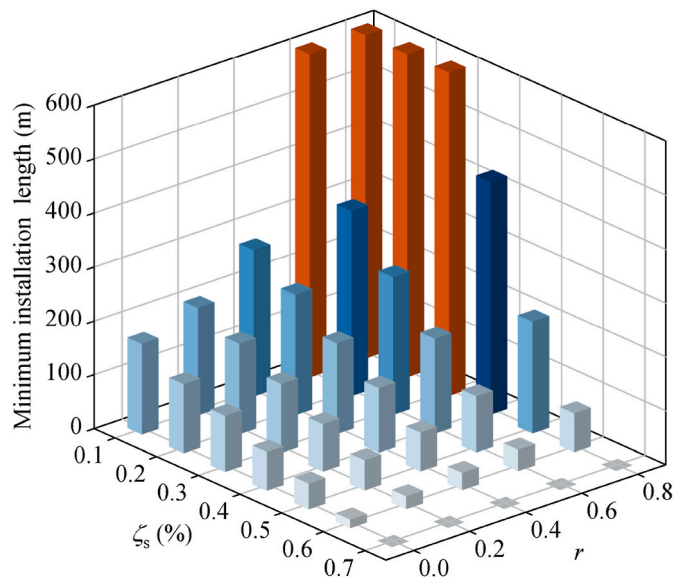


Fig. 16. Minimum required installation length of aerodynamic countermeasures to control multi-mode VIV with different structural damping ratios (ζ_s) and scale factors (r). The orange data means the multi-mode VIV cannot be simultaneously prevented even though the aerodynamic countermeasures are installed along the entire span of bridge deck.

will inevitably affect the optimized result, and it is necessary to analyse the sensitivity. To facilitate the discussion, the aerodynamic parameter $Y_{1,A}$ of CsA remains unchanged, while a scale factor r is introduced to amplify $Y_{1,B}$ of CsB, i.e., $Y'_{1,B} = rY_{1,A} + (1-r)Y_{1,B}$. According to Fig. 9, the positive aerodynamic damping acting on CsB will decrease to negative damping with the increase of r , which implies a reduction in the control effect of the baffle per unit length.

By conducting global symmetric optimization with different values of structural damping ratio ζ_s and scale factor r , the minimum required baffle installation length to prevent VIV in the 4 target modes has been determined. Fig. 16 shows the optimized results, which exhibit high sensitivity to both the structural and aerodynamic parameters. The minimum required length of baffles remarkably decreases as the structural damping ratio increases. As a result, it is unsafe to apply a layout scheme optimized under high structural damping condition to the bridge with lower damping. Given that the structural damping may gradually decrease with the degradation of bridge, it is recommended to conduct the optimization under relatively low structural damping condition. On the other hand, if the control effect of the baffle per unit length is reduced by increasing the scale factor r , this countermeasure needs to be installed over a larger length along the span. Too weak control effect of the countermeasure per unit length will lead to the failure of optimization (e.g., the orange results shown in Fig. 16). Therefore, it is worthwhile to explore a relatively effective aerodynamic countermeasure through the wind tunnel tests.

5. Conclusions

This study proposes a method to achieve an economical control scheme for multi-mode VIV by optimizing the spanwise layout of aerodynamic countermeasures on bridge deck. An improved mode-by-mode 3-D VIV analysis approach based on the generalized polynomial model is developed to calculate the VIV responses of the bridge with different spanwise layouts of aerodynamic countermeasures. The GA is then employed to search for the most economical layout scheme with constraints of limited multi-mode VIV responses. Two alternative types of search spaces are provided for GA: a global search space, and a reduced search space based on the theoretically optimized countermeasure

layout schemes for single-mode VIV control. The feasibility of this method is validated through finite element simulation and wind tunnel tests on a long-span suspension bridge. The major conclusions are summarized as follows:

- (1) The optimized layout scheme for single-mode or multi-mode VIV control achieves a notable reduction in the total installation length of aerodynamic countermeasures. Optimization with the global search space yields the most economical layout scheme, while the reduced search space offers a satisfactory scheme with much higher search efficiency.
- (2) Considering the practical construction requirement of real bridges, the aerodynamic countermeasures are installed in discrete segments along the bridge deck in the optimization. As a result, the proposed method may yield an asymmetric optimized countermeasure layout even for a symmetric bridge. To obtain a symmetric optimized layout scheme, additional constraints need to be introduced.
- (3) The proposed method may yield multiple optimum countermeasure layout schemes with the same objective function value. Any of these schemes can be chosen for practical application, because their control effects for VIV are relatively similar.
- (4) The optimized result of the proposed method is sensitive to structural and aerodynamic parameters. It is recommended to find an effective aerodynamic countermeasure for the bridge deck in advance and conduct the spanwise layout optimization of the countermeasures under relatively low structural damping condition.

CRediT authorship contribution statement

Hao Sun: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Software, Visualization, Writing – original draft. **Le-Dong Zhu:** Conceptualization, Funding acquisition, Resources, Supervision, Writing – review & editing. **Zhong-Xu Tan:** Methodology, Validation, Writing – review & editing, Funding acquisition. **Qing Zhu:** Funding acquisition, Methodology, Supervision. **Xiao-Liang Meng:** Investigation, Validation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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